

FELIPE RIBEIRO

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EDUCATION

University at Buffalo, State University of New York

Bachelor of Science in Mechanical Engineering

Buffalo, NY

Expected May 2027

Relevant Coursework: Mechanics of Solids, Statics, Dynamics, Mechanics of Materials, Machines & Mechanisms, Finite Element Analysis, Control Systems, Manufacturing Processes, Thermodynamics, Heat Transfer

TECHNICAL SKILLS

Robotics & Embedded: Arduino, ESP32, Raspberry Pi, PID & Kalman filtering, Gimbal systems

Programming: C++, MATLAB, Python, Excel

CAD & Analysis: Fusion 360, SolidWorks, AutoCAD, Prusa Slicer, FEA

ENGINEERING EXPERIENCE

UB ASME — IAM3D 2026 Rover Team

Buffalo, NY

Electronics Lead

August 2025 – Present

- Leading electronics integration for a 3D printed rover competition, evaluated on payload delivery effectiveness against competing colleges.
- Designed and wired a multi-battery power system, increasing rover motor performance by ~25% while regulating voltage for 6.25–8.25V high-torque servos.
- Programmed Arduino and ESP32 to translate RC receiver signals to ESC-driven DC motors, servos, relays, and onboard subsystems.
- Built and integrated a servo-actuated excavator scoop system with FPV camera support, laser alignment assistance, and custom 3D-printed protective enclosures.
- Competed with the IAM3D rover team at the EFX engineering competition (University of Central Florida).

Self-Balancing Inverted Pendulum Robot

Buffalo, NY

Personal Project

February 2026

- Built a 3D-printed two-wheel inverted pendulum robot using (Arduino, MPU-6050, DRV8833), cutting balance sway from $+8^\circ/-6^\circ$ to a steady $\pm 4.5^\circ$ by adding mass to raise inertia and rigidly mounting the IMU to reduce tilt-estimate noise.
- Fused accelerometer and gyroscope data via a two-state Kalman filter feeding a PID loop (K_p 90, K_i 0.2, K_d 6) for real-time balance and disturbance rejection.
- Diagnosed mismatched startup thresholds between the two TT gearmotors; measured each motor's activation point and used the `map()` function to scale PID output across both motors' usable PWM range, giving the controller finer resolution and better wheel response.

2-Axis Gimbal System

Buffalo, NY

Personal Project

March 2026

- Modeled and 3D-printed a 2-axis gimbal system for an FPV camera using servo motors controlled via remote control.
- Implemented filtering (EMA) and deadband handling to optimize performance and allow the gimbal to be controlled smoothly.
- Designed a custom gimbal frame with optimized geometry for low rotational inertia and balanced center of mass.

Servo-Actuated 2-DOF Excavator Payload Mechanism

Buffalo, NY

Personal Project

January 2026

- Designed a fully 3D-printed excavator-style payload system in Fusion 360 with two actuated joints (elbow, wrist) for controlled object manipulation.
- Created 3D-printed bearings and gears for stabilized movement at each joint; integrated high-torque servos with bearing-supported pivot points to reduce friction.

PROFESSIONAL EXPERIENCE

Infusion by Castles

New York, NY

Advertising Intern — Client Operations & Office Design

August 2022 – September 2024

- Managed and organized direct mail campaign files across multiple quarters, categorizing assets by market segment (International, AA, HS) for client access and review.
- Communicated directly with clients on campaign status, revisions, and delivery timelines, ensuring alignment between creative, production, and stakeholder teams.
- Led office layout redesign during company relocation, developing spatial plans and coordinating design decisions to support workflow efficiency and brand alignment.

ADDITIONAL INFORMATION

Activities & Interests: Fitness & bodybuilding, automotive restoration, hands-on fabrication, Investing